

Online Retail SQL Analytics Report

Executive Business Analytics Report

Tooling: Google BigQuery (Standard SQL)

Dataset Revenue: £8,797,499

Total Orders: 47,939

Total Customers: 4,338

Average Order Value (AOV): £476.06

1. Executive Summary

This project delivers a full-scale SQL analytics solution on an e-commerce retail dataset using Google BigQuery.

From 47,939 transactions across 4,338 customers, the business generated:

£8.8 million in revenue

The analysis uncovered:

- Strong seasonality with peak revenue of £1.14M in November 2011
- Revenue concentration in a dominant product category (£6.52M from “Other”)
- A high repeat customer rate of 65.4%
- Revenue heavily concentrated among a small set of high-value customers
- Top 10 products contributing £853,718 (~9.7% of total revenue)

This project demonstrates structured SQL modelling, dimensional design, and executive-level commercial interpretation.

2. Revenue Performance & Seasonality

Monthly Revenue Trend Highlights

Month	Revenue
Sep 2011	£943,114
Oct 2011	£1,024,240
Nov 2011	£1,140,650 (Peak)
Dec 2011	£512,831

Key Observations

- Revenue more than doubled from February 2011 (£442,919) to November 2011 (£1,140,650).
- Q4 shows strong seasonal uplift.
- November alone contributed ~13% of annual revenue.

Business Implication

- Strong holiday-driven demand.
- Inventory and marketing spend should be strategically increased ahead of Q4.
- December drop suggests possible post-peak supply or demand normalization.

3. Revenue by Product Category

Category	Revenue	Orders
Others (Miscellaneous)	£6,522,422	17,799
Accessories	£874,013	8,052
Home Décor	£578,761	8,066
Stationery	£525,769	7,449

Strategic Insight

- “Other” accounts for 74% of total revenue.
- The top four categories collectively generate over 96% of revenue.
- “Kids” category contributes only £31,548 (0.36% of revenue).

Business Recommendation

- Reclassify “Other” category to gain clearer product insights.
- Expand high-performing categories.
- Reassess viability of low-performing segments.

4. Product-Level Revenue

Top 10 Products Revenue

Total Revenue (Top 10): £853,718

Total Units Sold (Top 10): 348,343

Top Product:

- PAPER CRAFT, LITTLE BIRDIE → £168,470 revenue, 80,995 units

Revenue Concentration

Top 10 products contribute:

$£853,718 \div £8,797,499 = 9.7\%$ of total revenue

Interpretation:

- Revenue shows Pareto-like distribution.
- A small number of SKUs materially influence performance.

Business Implication:

- Prioritise stock availability for top 10 SKUs.
- Use top SKUs for bundle strategy to life slower-moving inventory.

5. Customer Behaviour Retention

Total Customers: 4,388

- Repeat Customers: 2,837
- One-Time Customers: 1,501
- Repeat Rate: 65.4%

Interpretation

Nearly two-thirds of customers are repeat buyers.

This is a strong retention signal.

Business Impact

- High ROI potential from loyalty programs.
- Repeat customers are likely driving revenue stability.
- Acquisition efforts should be paired with retention optimisation.

6. Average Order Value (AOV)

AOV: £476.06

This is a relatively high basket size for retail.

Interpretation

- Customers purchase in bulk or higher-value combinations.
- Cross-selling and bundling strategies are already partially effective.

Growth Opportunity

- Structured upsell and recommendation systems could lift AOV beyond £500.

7. Customer Lifetime Value (LTV) Distribution

Top 10 Highest LTV Customers

Highest Revenue Customer:

Customer 14646 → £279,138 (72 orders)

Second Highest:

Customer 18102 → £259,657 (60 orders)

Notably:

Customer 14911 generated £143,711 across 201 orders

Observation

Some high revenue customers place many smaller orders, others place fewer but high-value orders.

Concentration Insight

Top 10 customers collectively generate:

£1,530,998

This represents:

~17.4% of total revenue

Business Implication

Revenue is significantly concentrated.

VIP-tier targeting could materially improve profitability.

8. Customers with Most Orders

Customer 12748 → 209 orders but only £33,049 revenue.

Customer 14911 → 201 orders and £143,711 revenue.

Insight

Order frequency does not always correlate with total revenue.

Some high-frequency customers generate low per-order revenue.

Recommendation

Segment by:

- High frequency / low value
- Low frequency / high value
- High frequency / high value

Each segment requires different engagement strategy.

9. Lowest Revenue Customers

Lowest recorded revenue:

£3.75 (single order)

Several customers generated less than £20 total revenue.

Insight

Long-tail customer base contributes minimal revenue.

Acquisition cost for these customers may exceed value.

10. Strategic Business Recommendations

1. VIP Segmentation Strategy

Top 10 customers generate ~17% of revenue.

Implement:

- Tiered loyalty benefits
- Personalised offers
- Early-access campaigns

2. Product Portfolio Optimisation

Break down the “Other” category to improve visibility and forecasting.

3. Seasonal Planning

Q4 revenue surge indicates strong seasonal dependence.

Optimise:

- Pre-holiday inventory procurement
- Targeted seasonal marketing

4. Customer Retention Focus

65.4% repeat rate is strong.

Retention investments likely yield higher ROI than aggressive acquisition.

5. AOV Growth Strategy

Implement bundle offers and dynamic pricing experiments to increase AOV beyond £476.

11. Technical Capabilities Demonstrated

- Star Schema Modelling
- Dimensional Design
- SQL Aggregations & Window Functions
- Revenue KPI Engineering
- Customer Segmentation Logic
- Business-Oriented Interpretation of Data
- Data Cleaning & Preprocessing Awareness

Final Positioning Statement

This project demonstrates the ability to:

- Translate raw transactional data into structured dimensional models
- Engineer scalable KPIs using SQL
- Quantify revenue concentration and customer value
- Deliver strategic insights aligned with executive decision-making

It reflects production-ready SQL and strong commercial and analytical thinking suitable for Data Analyst, BI Analyst, and Business Analyst roles.